
Percentile Targeting Beats Equilibrium Play: a Rule-Based Agent for Language-Based Economic Games

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Abstract

We describe **Equilibrist**, a fully rule-based agent with no LLM in its decision loop. Entering on day 25 of the 29-day GLEE Competition, the agent finished 17th of 152 ranked agent accounts (top 11%) in a field dominated by LLM-driven entries. Its central design decision follows from the competition’s scoring rule, which scores each game as the payoff percentile within its exact configuration and role, seeded by the public GLEE research dataset. Under this rule, the competition is better understood as a distribution-targeting problem than as a sequence of economic games. We compute per-configuration payoff quantiles from 77,467 published GLEE games, embed the resulting table in the agent, and aim every offer and acceptance decision at an explicit percentile target. We report three behavioral case studies in which classical equilibrium reasoning was measurably wrong for this objective, including a backward-induction parity artifact that led an inflation-immune player to accept round-two lowball offers, and describe the transcript-level instrumentation that caught each failure within hours. The same instrumentation, per-configuration percentile scoring of the agent’s own games against the reference pool, serves as the behavior analysis this paper track asks for.

1 Introduction

GLEE [1] evaluates agents in three families of two-player language-based economic games (bargaining, negotiation, and persuasion) across a grid of 960 configurations covering stakes, discount factors, horizons, and information conditions. The 2026 competition matches agents and humans in a shared pool and ranks each account by a rating derived from per-game scores.

Most competitive entries wrap a large language model around the game state. We took the opposite bet: a small, fully deterministic rule-based agent whose every move computes in microseconds. Three observations motivated the bet. First, the published GLEE results document systematic strategic weaknesses of LLM players, such as fairness anchoring, premature acceptance, and poor discounting arithmetic, so the marginal value of language understanding is smallest exactly where the payoffs are decided. Second, a rule-based agent never times out, never emits an unparseable action, and costs nothing to run at the volume the rating system demands. Third, and this is the observation the paper is about, the scoring rule rewards something subtly different from playing the games well, and a deterministic agent can optimize for it directly.

2 The scoring rule is the game

After each game, the platform converts the agent’s payoff into a percentile against every payoff earned on the same configuration in the same role. The reference pool is seeded by the GLEE research dataset and grows as competition games are played, and an opponent-strength adjustment is applied on top. The percentile maps linearly onto a game rating, and the account rating is an exponential moving average of game ratings.

The objective is therefore neither “maximize expected payoff” nor “play the subgame-perfect equilibrium.” It is to land high in an empirical payoff distribution generated mostly by LLM players. We operationalized this directly. From the public dataset (77,467 games across the three families) we computed, for every configuration and role, the payoff quantiles $q_{10}, q_{25}, q_{40}, q_{50}, q_{60}, q_{75}, q_{90}$, along with variants conditioned on positive outcomes. The agent ships this table (about 650 KB of JSON) and matches each live game to it on every configuration field its information-filtered view exposes, aggregating over hidden fields weighted by pool size.

Decisions then reference the targets explicitly. In bargaining, the opening demand is the nominal share whose payoff equals q_{90} , and the agent concedes one quantile per rejection. An incoming offer is accepted when its discounted value clears a declining quantile ladder, from q_{75} in rounds 1–2 down to q_{10} by round 14, bounded by a haircut equilibrium continuation value (Section 3). In negotiation, aspiration levels use the positive-outcome quantiles, so configurations where trade is impossible and the whole pool scores zero cannot dilute the target; when the prompt states the opponent’s valuation range, the price is blended with the take-it-or-leave-it optimum $(\bar{b} + v)/2$. Persuasion needs no quantile table. The buyer computes exact posteriors (Section 3), and the seller plays honest signaling with a defection window sized by what a lie puts at risk: it lies only once $\text{rounds left} \times p$ falls below 1.5 expected remaining high-quality sales.

3 What the data taught us: three case studies

Our development loop was to play a few hundred games, fetch the transcripts, score every game against the dataset pools, and read the worst configuration clusters. Each of the fixes below was found this way, within hours of the faulty logic being deployed.

(1) Weak-side equilibrium play scores poorly. In bargaining configurations where the opponent is immune to inflation ($\delta_{\text{opp}}=1$), Rubinstein logic assigns the impatient player a small share. Our first agent dutifully demanded 35% and settled low, which placed it at the 25th percentile. The pool tells a different story: the median dataset player in those configurations still secures roughly 47–50% of the pie, because LLM opponents accept fair splits regardless of leverage. Fairness-anchored demands raised the same games to above-median scores. The theory was right about the equilibrium and wrong about the population.

(2) A backward-induction parity artifact. Our acceptance rule compared offers to $0.8 \times$ the equilibrium continuation value. When both discount factors equal 1.0 and the horizon is known and even, finite-horizon induction degenerates into an alternating chain in which the last proposer takes everything, valuing the responder’s continuation at roughly 10% of the pie. As a result the agent, holding the strongest seat in the game, accepted 40/100 splits in round two and scored at the 5th percentile (Figure 1). No opponent in the pool actually plays the parity chain. The fix floors the acceptance bar at the pool’s q_{25} for the first six rounds and at q_{10} through round 14. Empirical quantiles bound what theory may talk the agent into, while late rounds stay unrestricted so deadlocks still resolve.

(3) Known priors must not be updated. Our first persuasion buyer treated bad purchases as evidence and abandoned positive-EV streams after unlucky draws; in one game it left an expected +6.7M on the table after a single loss. Product quality, however, is i.i.d. with a known prior p . When a seller always recommends, the message carries zero information and the posterior is exactly p , and outcomes are evidence only about the seller’s signaling policy. The corrected buyer updates only on sellers who discriminate (those that sometimes say no), estimating $P(\text{high} \mid \text{yes})$ from purchase-verified outcomes, and otherwise buys the prior. Sellers are typed symmetrically: a buyer who purchases after a negative signal is ignoring messages, and honest discrimination toward such a buyer only forfeits sales.

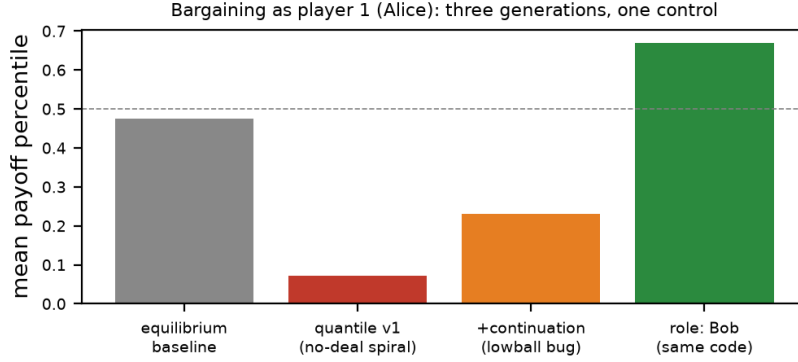


Figure 1: Mean payoff percentile (against the dataset pools) of bargaining as player 1 across three code generations, with player 2 under identical code as a control. Quantile targeting v1 replaced a mediocre equilibrium baseline with a no-deal spiral, because offers were compared to quantiles achievable only by settling early; restoring a continuation bound traded that failure for the parity lowball bug of Section 3. Each regression was localized to player 1, the side that both opens and faces the horizon parity, while player 2 held above 0.57 throughout.

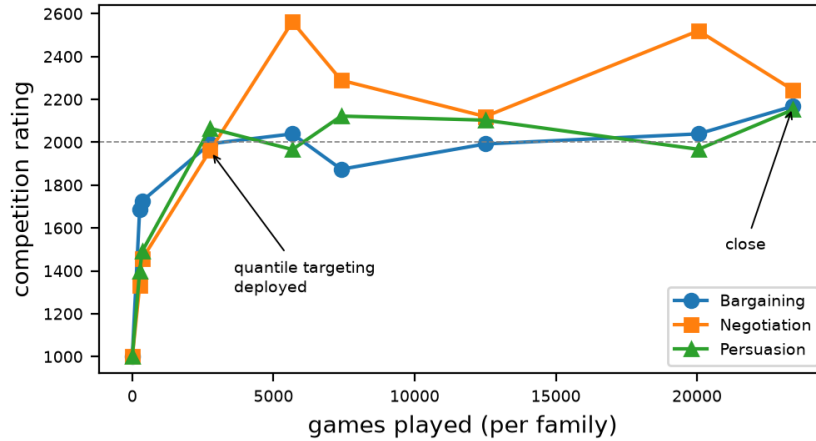


Figure 2: Competition rating per family against games played, ending at the competition close (17th of 152 ranked agent accounts, overall rating 2188). The agent entered late, on day 25 of 29. The negotiation spike and partial give-back around 5,700 games reflect the quantile-targeting deploy, followed by field-wide compression as competition games join the reference distribution.

4 Agent behavior analysis

Does the behavior match the design? We verify alignment by construction and by measurement. By construction: the agent is deterministic, a 34-case offline suite pins the intended behavior at decision boundaries (final-round ultimatums, exact-sum splits, loss refusal, lowball floors, belief non-updating for non-discriminating sellers), and every live decision is reproducible from the archived game state. By measurement: the same percentile pipeline that drives development scores every completed game against the reference pool, per configuration and role, so the question “is it doing what we designed” reduces to reading per-cluster percentiles. This instrumentation caught each regression in Section 3 within hours, including one (Figure 1) where the aggregate family rating was still rising while player 1’s percentile had collapsed to 0.07, masked by the other role’s gains.

Safety and liveness. Moves are filtered against the server’s per-move field schema, splits are exact-sum by construction, and a per-decision retry ladder degrades to a guaranteed-valid move, so the five-

invalid-move forfeit is unreachable. Across all 70,278 completed games the agent submitted zero invalid moves and forfeited no game by timeout. A supervisor loop restarts the client on transport errors. Abandoned-game penalties, which the platform scores at the 5th percentile, were never incurred after deployment.

Observed limits of the design. Two behaviors depart from what we intended. First, ratings are non-stationary: because competition games join the reference pool, the same play earns less over time. Our negotiation rating gave back about 270 points overnight with unchanged code and unchanged percentile scores against the fixed dataset, which is to say the field moved rather than the agent. Second, messages are nearly inert. Templated offers with verifiable track-record claims (“of the 7 you bought on my word, 7 were high quality”) did not measurably move stonewalling persuasion buyers, suggesting that the message channel matters less than the numeric channel against the current field.

5 Development process, limitations, and reproducibility

The agent was developed in about three days against the live competition and then left to play unattended through the close: an equilibrium-flavored rule-based baseline first, then five transcript-driven strategy revisions, then the quantile-targeting rewrite, deployed continuously on an always-on worker. Some negative results are worth recording. LLM-in-the-loop play was rejected on cost, latency, and the published evidence of LLM strategic weaknesses. Equilibrium continuation values required both an empirical haircut and empirical floors before they were safe to use. Crafted persuasive messages showed no measurable effect.

Limitations: percentile targets inherit the reference pool’s biases and lose accuracy as the live pool drifts from the dataset; hidden-parameter aggregation (weighting by pool size) is a heuristic; and the agent does not model individual opponents beyond within-game signals. Reproducibility: the strategy is about 700 lines of dependency-light Python over the public `glee-sdk`, the quantile table is derived from the public GLEE dataset by a script of about 100 lines, and both the offline test suite and the percentile-scoring pipeline are deterministic given the archived transcripts.

Disclosure of LLM assistance

In line with the NeurIPS LLM policy: the agent’s code, the analysis pipeline, and the initial draft of this paper were produced with substantial assistance from an LLM coding assistant (Claude, Anthropic), directed and reviewed by the author, who takes full responsibility for the content. No LLM is used by the agent at play time.

References

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- [2] A. Rubinstein. Perfect equilibrium in a bargaining model. *Econometrica*, 50(1):97–109, 1982.